

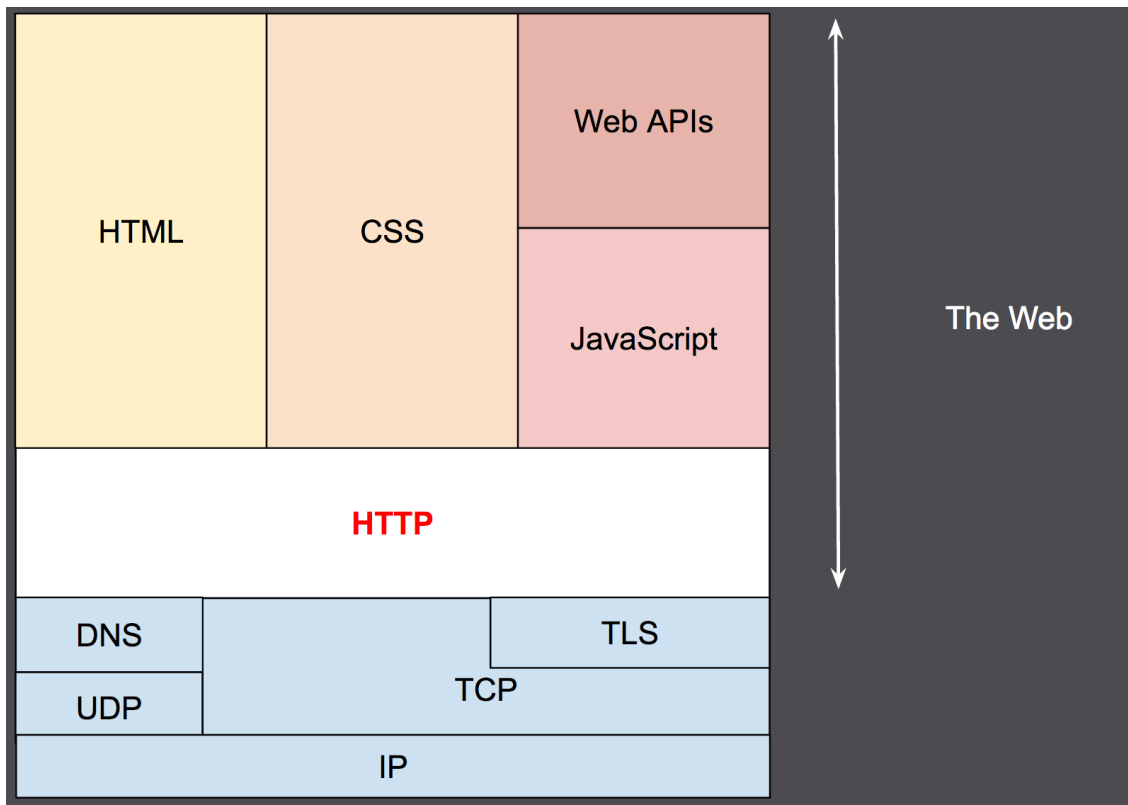
Protocolos

http://

Desarrollo de Aplicaciones para Internet

HTTP

un protocolo para comunicar
aplicaciones por internet

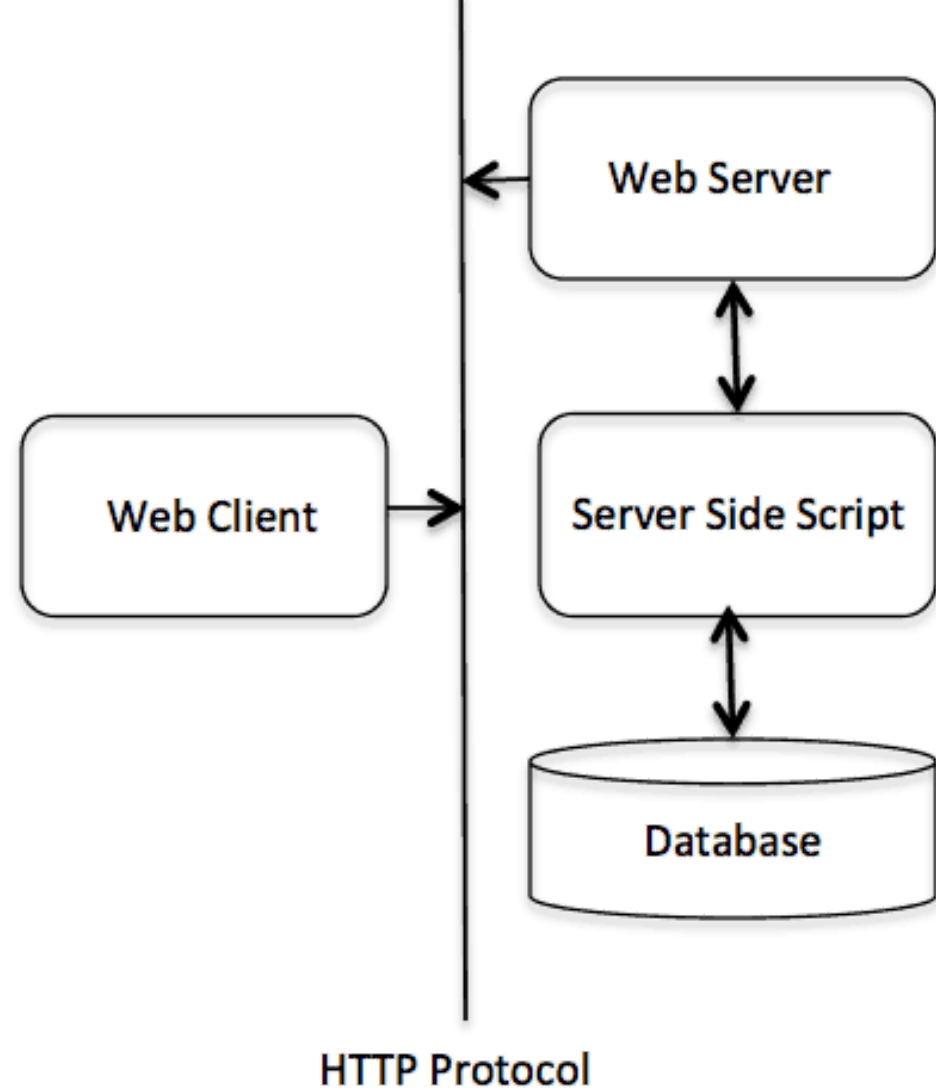


Las aplicaciones web están
distribuidas en dos:

en un 99% de los casos

- Cliente
- Servidor

Comunicadas con **HTTP**



Cliente

Es **proactivo**,
inicia la comunicación

Manda **requests**

Se encarga de la
interface de usuario

Servidor

Es **reactivo**, está esperando los
requerimientos de los clientes

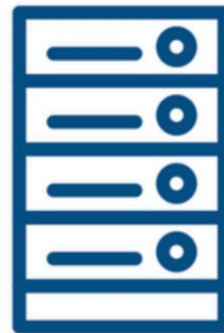
Responde con **responses**

Lógica de la aplicación,
comunicación con la Base de Datos



Front End

- Markup and web languages such as HTML, CSS and Javascript
- Asynchronous requests and Ajax
- Specialized web editing software
- Image editing
- Accessibility
- Cross-browser issues
- Search engine optimisation

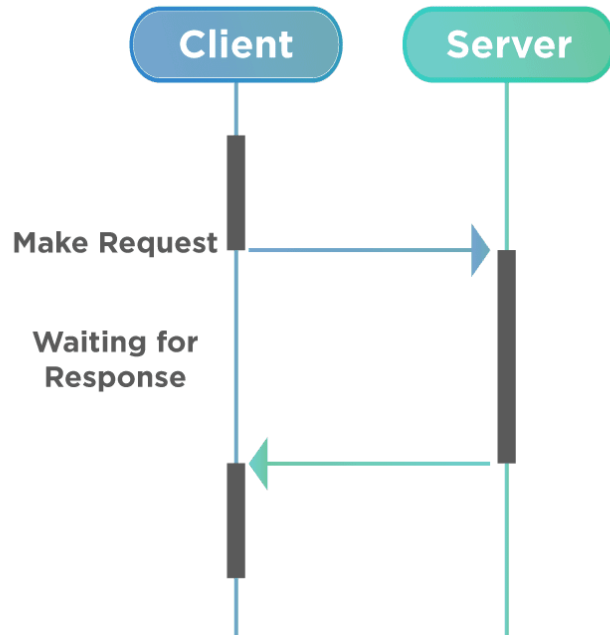


Back End

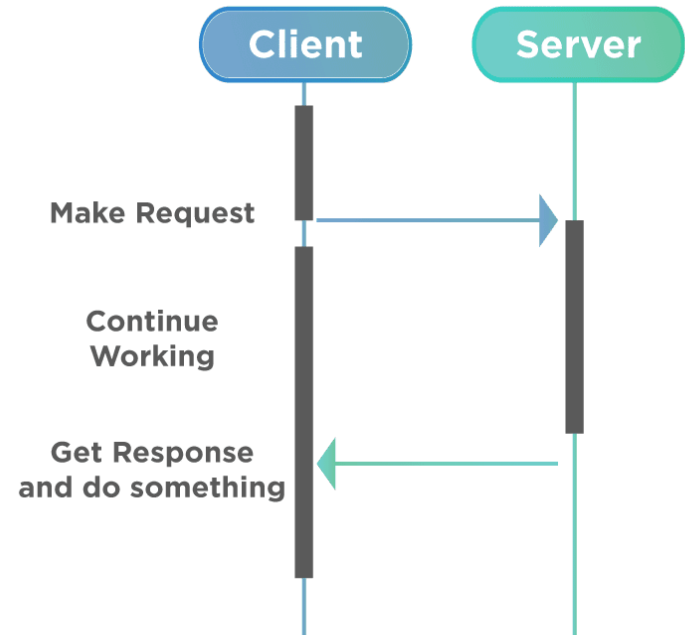
- Programming and scripting such as Python, Ruby and/or Perl
- Server architecture
- Database administration
- Scalability
- Security
- Data transformation
- Backup

La comunicación puede ser síncrona o asíncrona

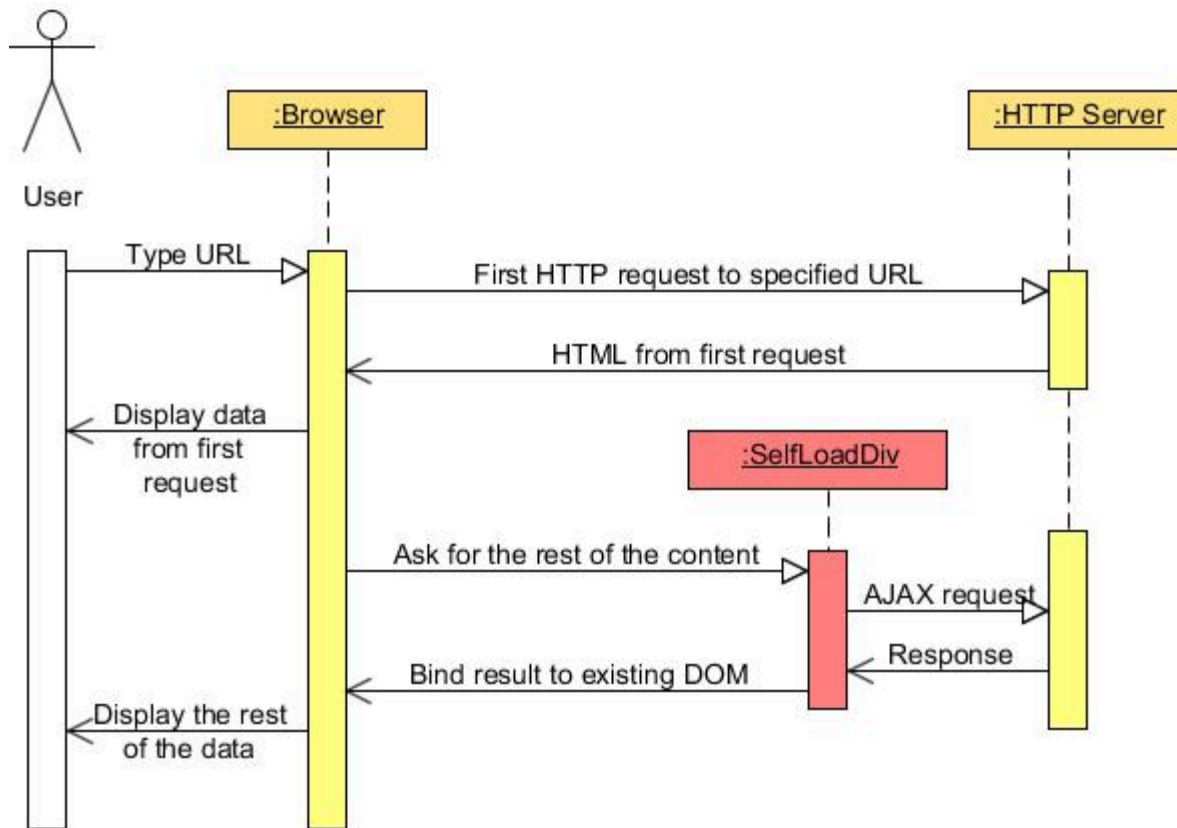
Synchronous



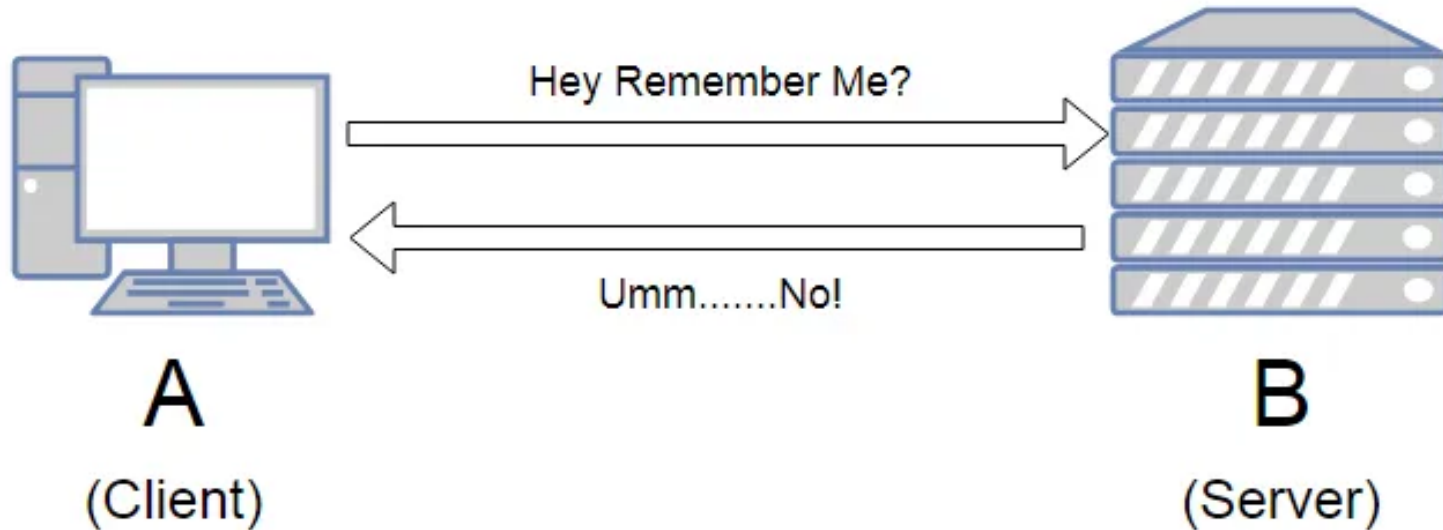
Asynchronous



Con llamadas asíncronas: AJAX, y fetch



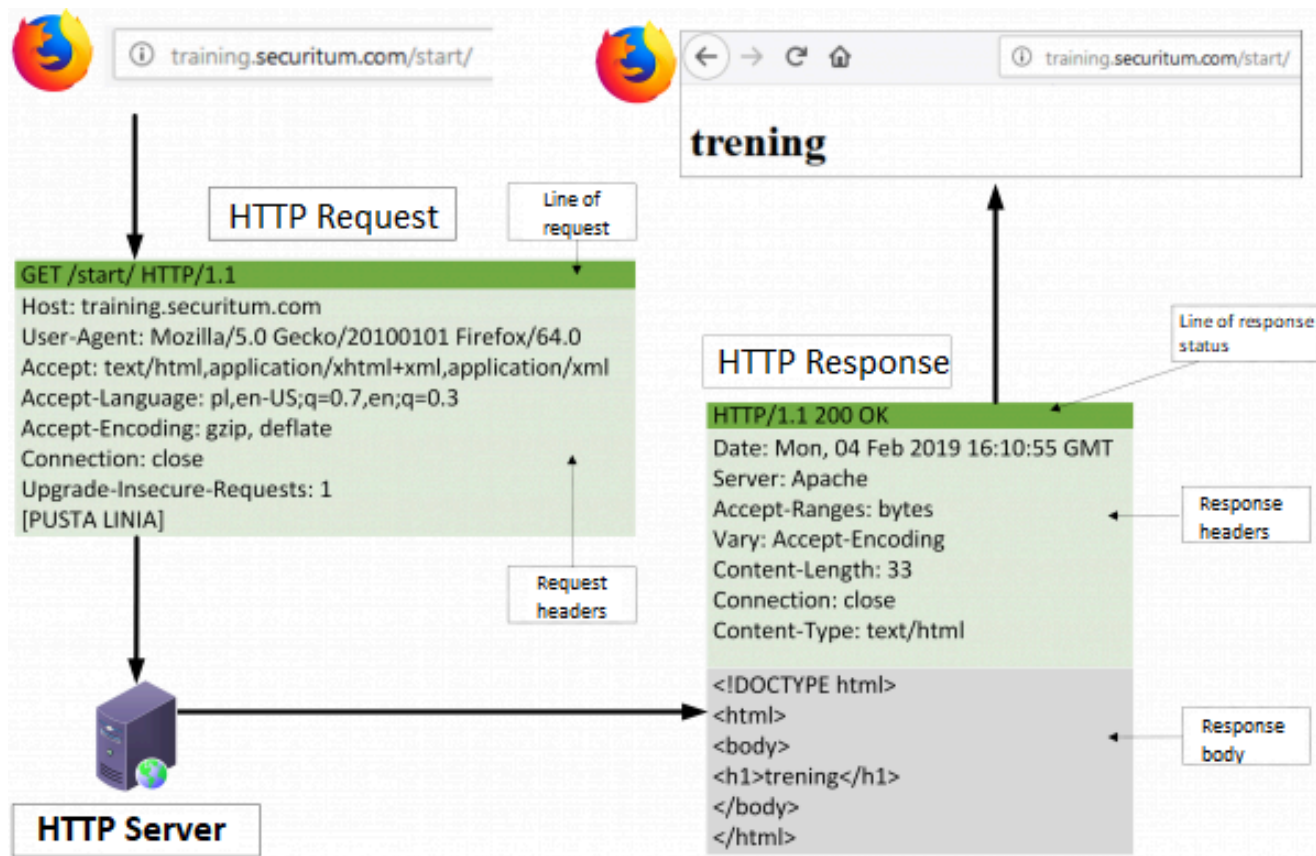
HTTP es un protocolo sin estado



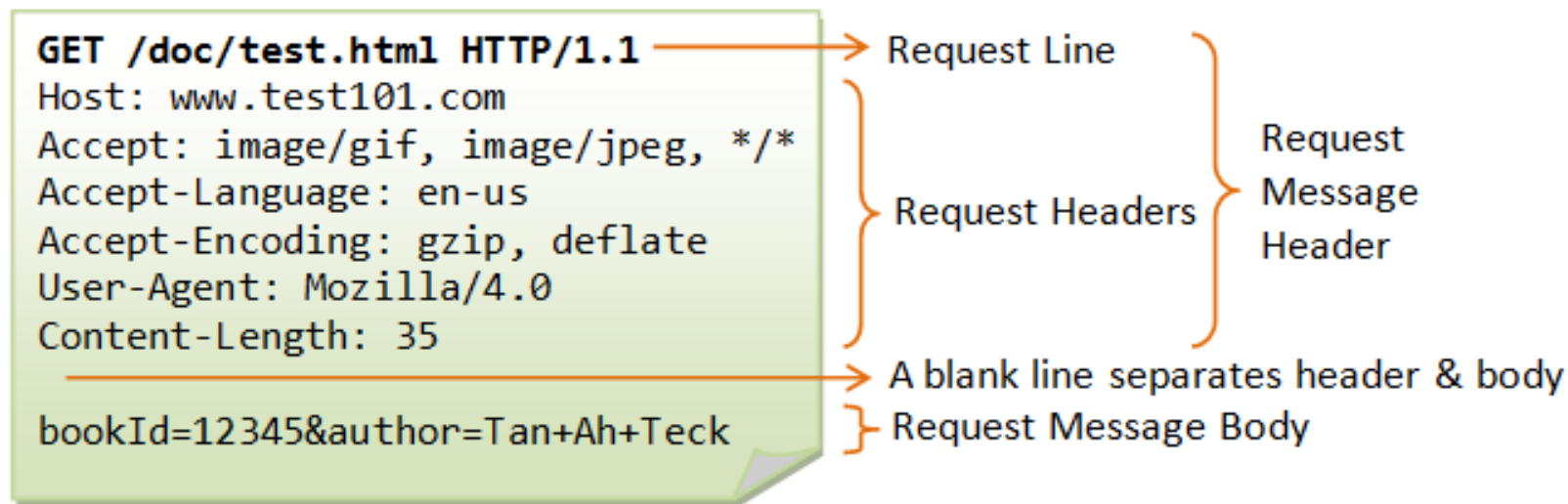
La respuesta no depende de ningún requerimiento anterior

y hay buenas razones

Http es un protocolo **ASCII**, con **requests** y **responses**



El request



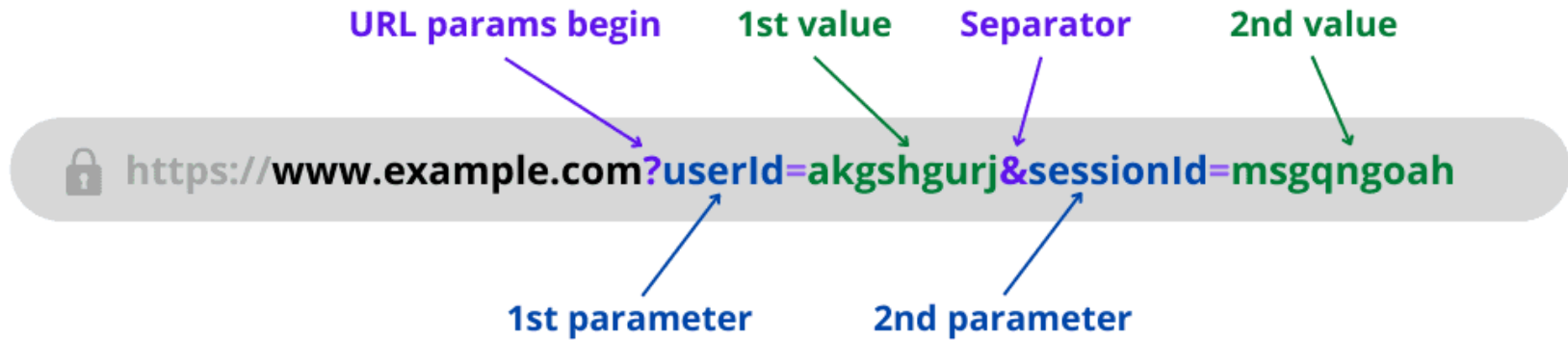
Request header

Verbos del http



Asociados a operaciones de CRUD cuando se usan en una API

Parámetros en el **GET**, en el URL



urlificados (HTTP es ASCII), incluso slug

Parámetros del **POST**, en el body

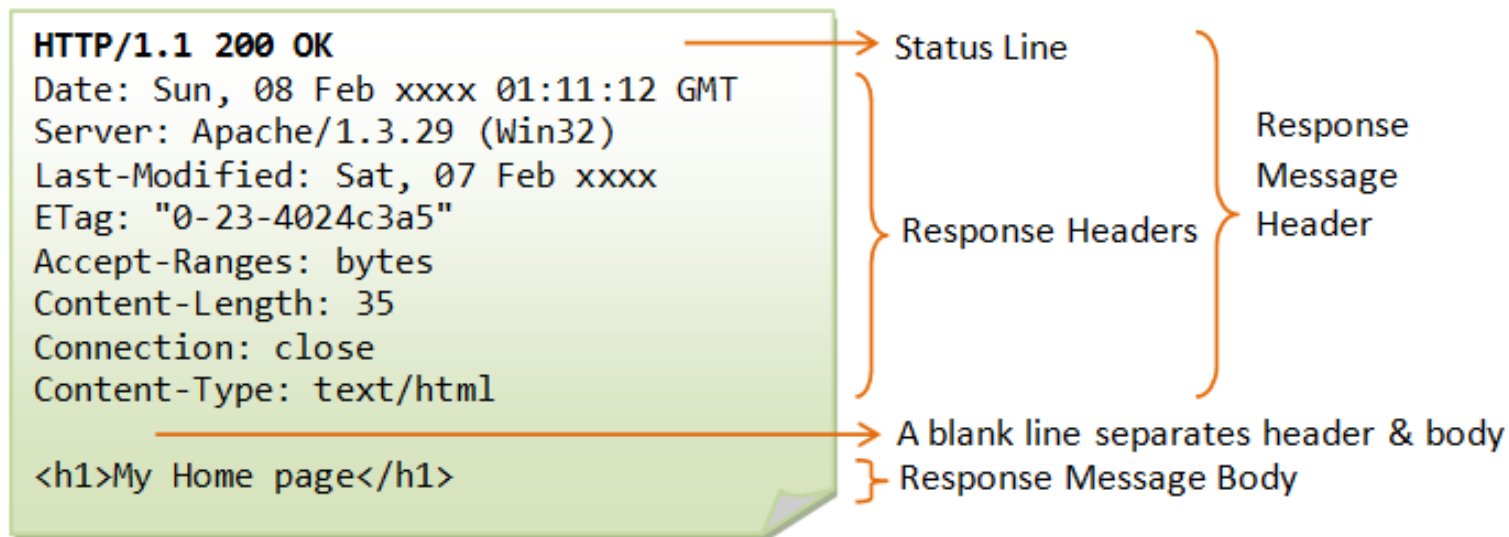
```
POST /repcion_formulario HTTP/1.1
Host: ...
...
variable_1=valor_1&variable_2=valor_2
```

cuando

```
<form action="repcion_formulario" method="post">
  <input name="variable_1">
  <input name="variable_2">
  <input type="submit" value="Enviar">
</form>
```

HTML Form Action: POST and GET

El response



Con status code

HTTP es sin estado, pero no sin sesiones, con cookies



Types of Cookies

Session Cookie

A session cookie is only active whilst the user is reading or navigating the website. When the user closes their web browser these cookies are usually removed.



Persistent cookie

A persistent cookie for a website exists on a users computer until a future date. For example the cookie expiry date could be set as 1 year, and each time a website is accessed over this period the website could access the cookie.



Http Only cookie

A Http Only cookie can only be used via HTTP or HTTPS, and therefore cannot be accessed by javascript. This reduces threat of session cookie theft via cross site scripting (XSS).



Secure cookie

A secure cookie can only be used via HTTPS. This ensures the cookie data is encrypted, reducing the expose to cookie theft via spying.

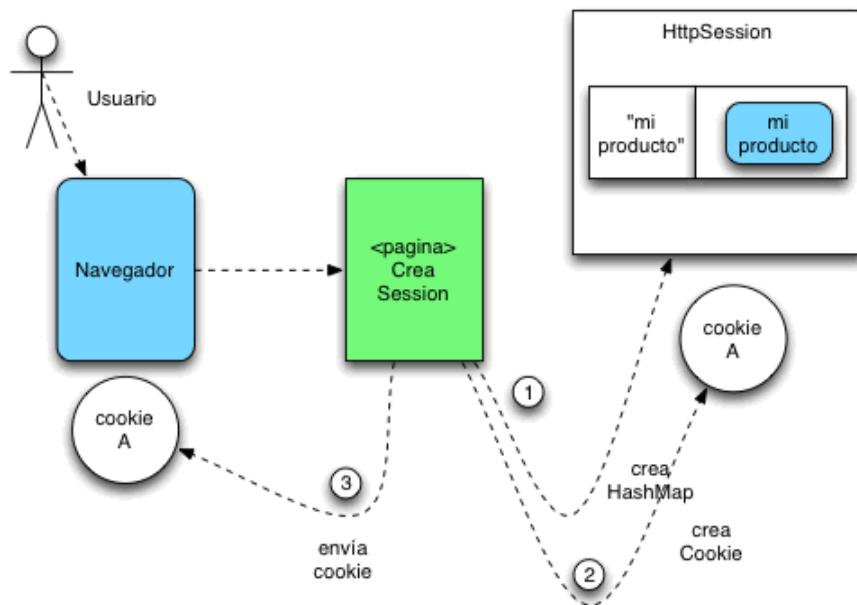


Third-party cookie

Third-party cookies are cookies set with domains, that are different from the one shown on the address bar. Privacy setting options in most modern browsers allow you to block third-party tracking cookies.



Se usan librerías de **sesiones** para facilitar el uso de cookies

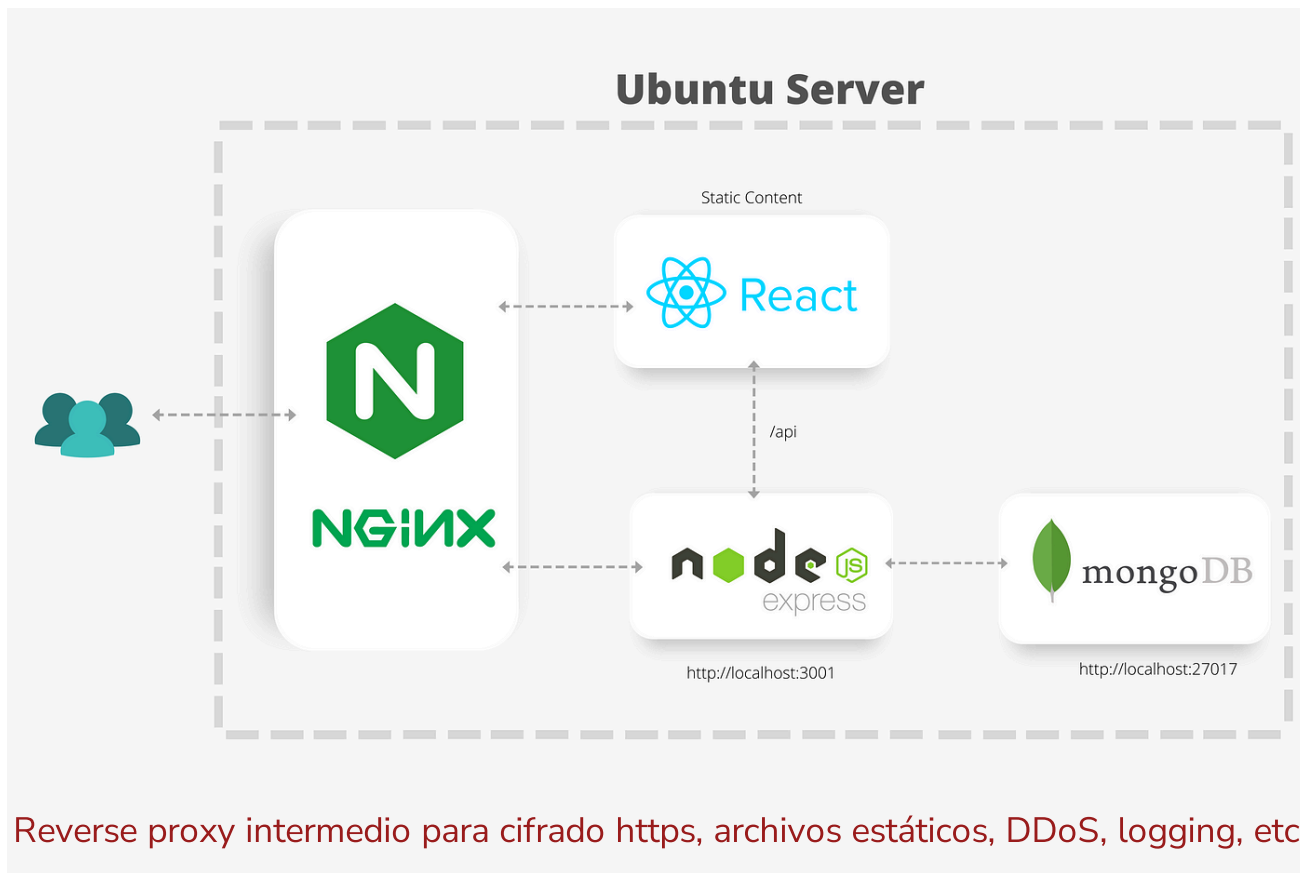


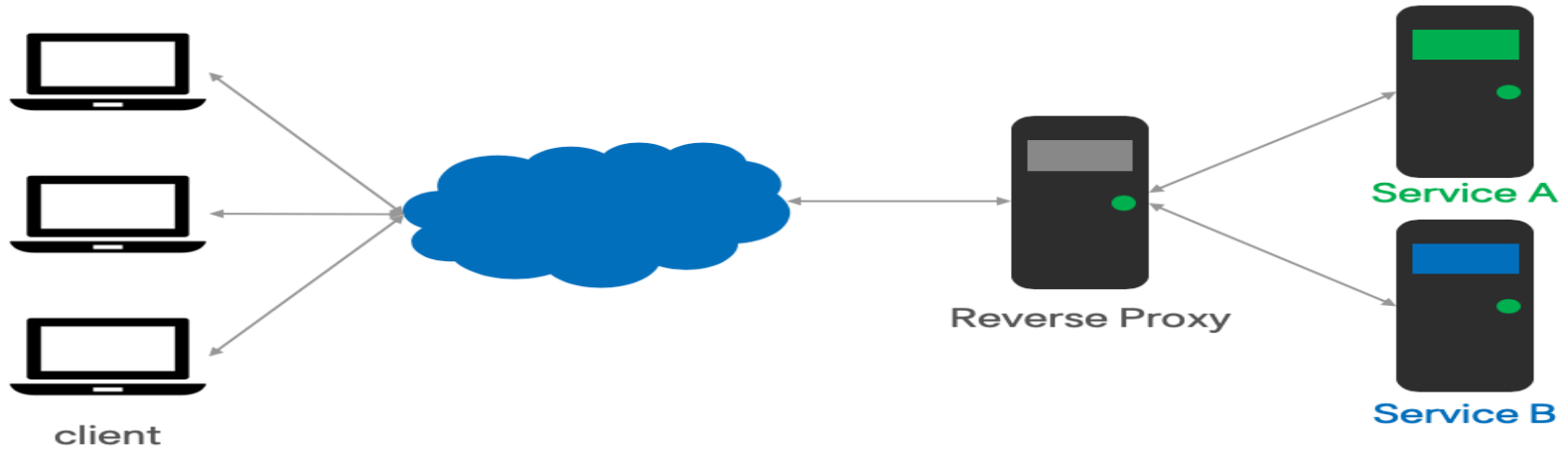
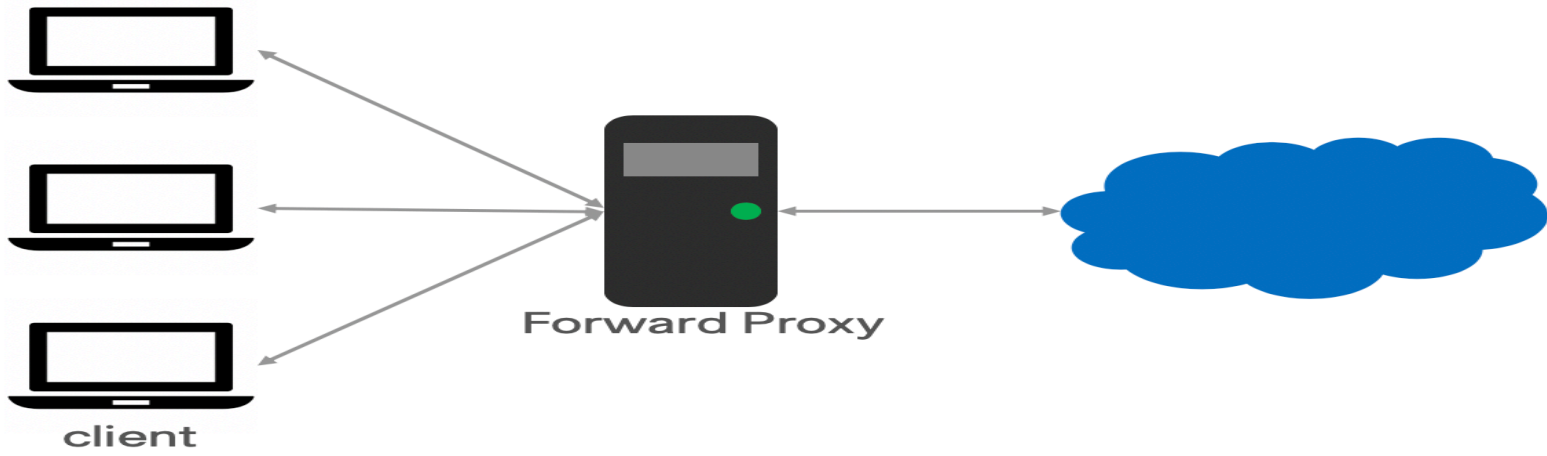
What are sessions? How do they work?

Evolución del protocolo

- HTTP 0.9
- HTTP 1.0
- HTTP 1.1
- HTTP 2

En producción



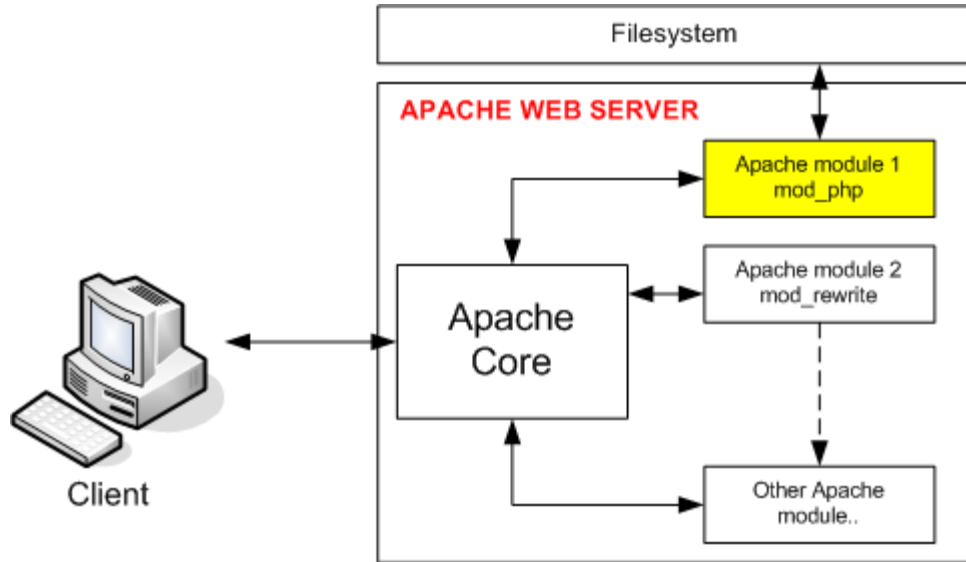




```
server {  
    listen      80;  
    server_name localhost;  
  
    location /static {                               /* archivos */  
        root    /var/www/static;  
    }  
  
    location /service_A {                           /* proxy inverso */  
        proxy_pass http://localhost:3000/;  
    }  
  
    location /service_B {  
        proxy_pass http://localhost:3001/;  
    }  
}
```

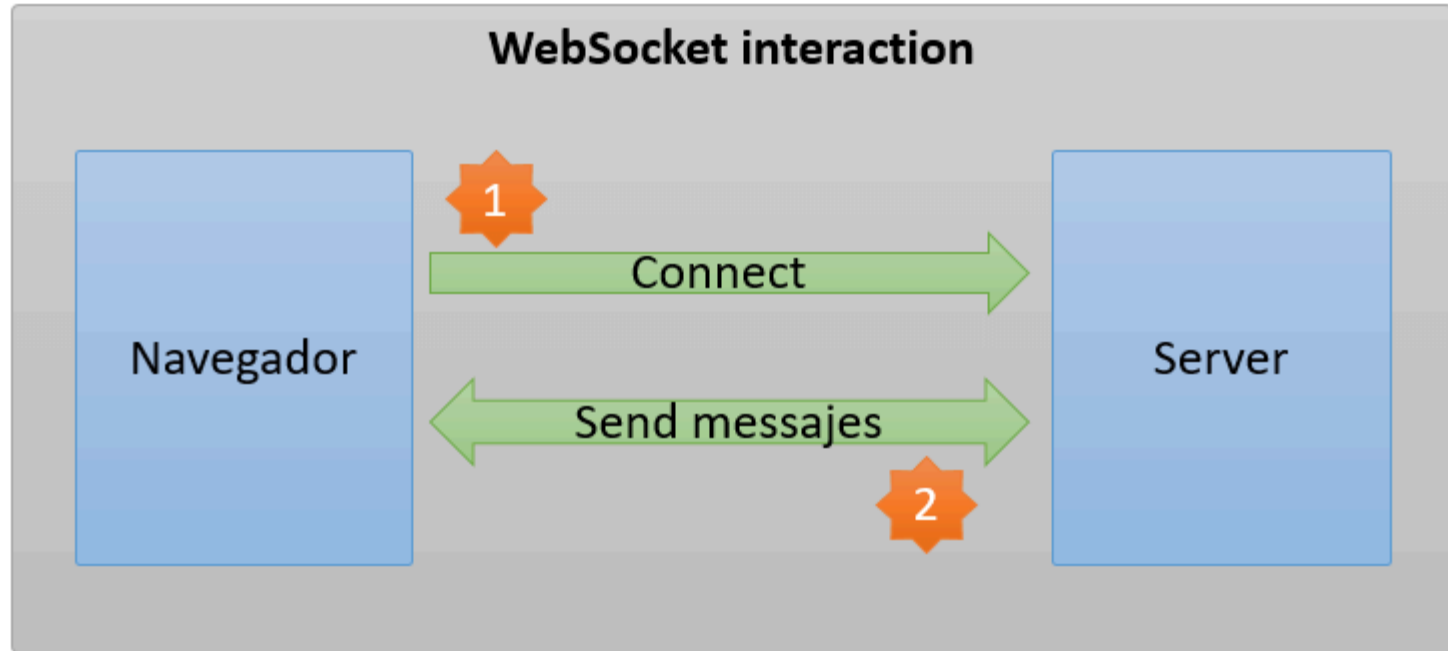
Configuración nginx

Distinto de como va Apache con mod_php

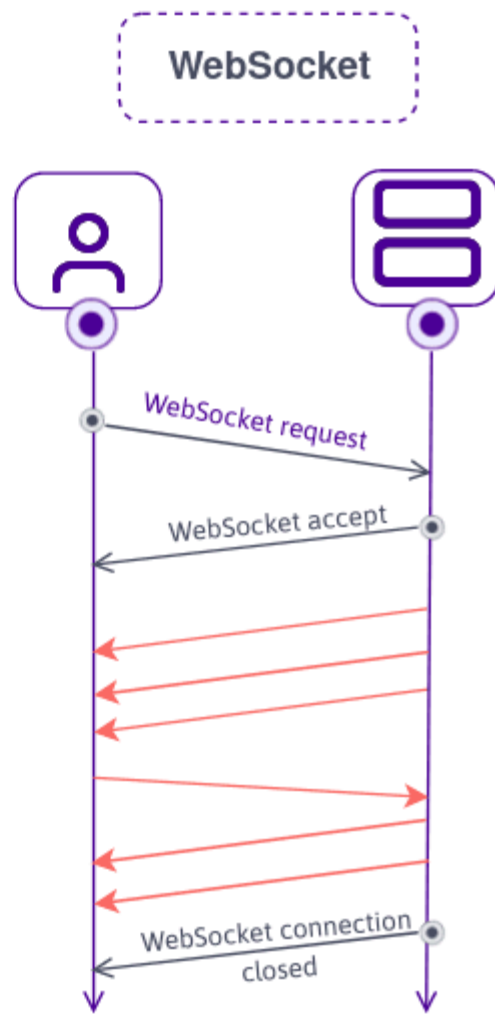
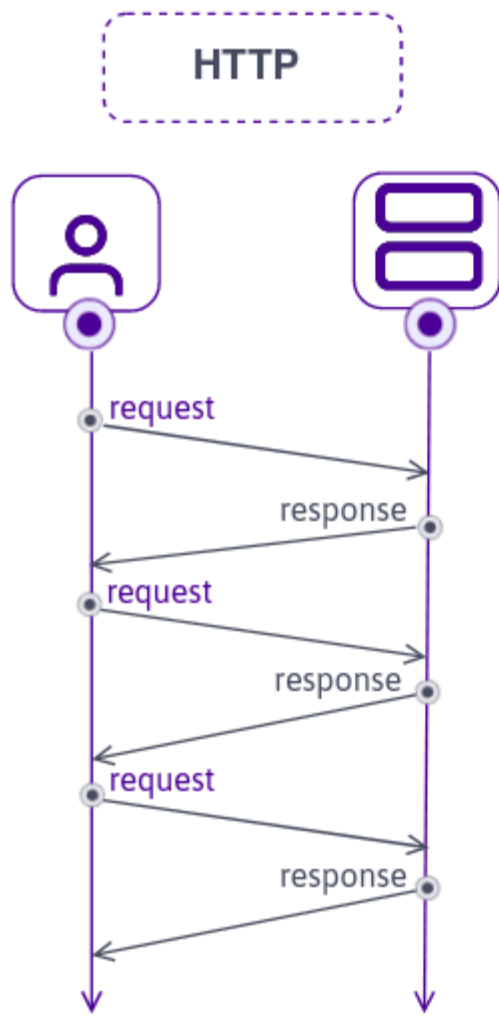


mod_php es un plugin

Full duplex con WebSocket



Una ampliación de HTTP para comunicación bidireccional



Handshake



Petición

```
GET /chat HTTP/1.1
Host: server.example.com
Upgrade: websocket
Connection: Upgrade
Sec-WebSocket-Key: x3JJHMbDL1EzLkh9GBhXDw==
Sec-WebSocket-Protocol: chat, superchat
Sec-WebSocket-Version: 13
Origin: http://example.com
```

Respuesta

```
HTTP/1.1 101 Switching Protocols
Upgrade: websocket
Connection: Upgrade
Sec-WebSocket-Accept: HSmrc0sMlYUkAGmm5OPpG2HaGWk=
Sec-WebSocket-Protocol: chat
```